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SYSTEMS DEVELOPMENT PROJECT

Module Name	**SYSTEMS DEVELOPMENT PROJECT**
Project Name	**MAKHANDA CONNECT SYSTEM**
Document Description	**NARRATIVE OVERVIEW**

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Mobile phone penetration in South Africa is high, with 124–127 million active cellular connections. Furthermore, over 90% of Internet users in South Africa own smartphones, and we are ranked as the number one country in the world for daily mobile Internet usage. Unsurprisingly, mobile phones are increasingly being used for government (especially at the local level) and citizen engagement and two-way communication.

To encourage and foster increased communication between the Makana local government and the citizens of Makhanda, a web-based platform called **Makhanda Connect** should be designed and developed as a central information and issue-reporting system for the citizens of Makhanda (a small peri-urban town in the Eastern Cape).

The system should combine a public landing page providing town-wide information (such as widespread power- or water-related problems) with dedicated ward-level spaces (a ward is a subdivision of a municipality created to allow for local-level representation within the municipality) where community members can log municipal service faults. Ward councillors can then manage, track, and report on these issues for submission to the local municipality.

The typical business processes that the system should cater to include:

- Facilitate account creation/registration of community members, ward councillors, municipal communications officers/local municipal managers, and system administrator accounts.
- Provide a view of general town information, events, safety notices, local weather, local dam levels, and the ability to browse ward summaries.
    - The general public should have access to at least a general landing page and ward-specific subpages (which provide ward-specific general information)
- Allow registered community members and ward councillors to create municipal service fault tickets for various municipal services (water, sanitation, electricity, road infrastructure, parks and recreation, etc) for their wards.
- Allow registered community members and ward councillors to track the progress of tickets for their wards.

- Allow ward councillors to manage and update tickets in their ward. They should be able to change a ticket's status from open to reported to resolved (resolved tickets are closed).
- Allow ward councillors to add notes to fault tickets in their ward.
- Allow ward councillors to generate ward-level summaries which are available via the system to the general public (including those without accounts).
- Allow municipal communications officers to add content, specifically community-wide or ward-level municipal notices.
- Allow system administrators to manage users (of all types), such as assigning/verifying roles to new account creations.
- Allow system administrators to manage content, and approve or update all community-wide notices.

System reports would include (but not limited to!):

- Service fault statistics per ward (daily, weekly, monthly, yearly, you decide).
- Service fault progress.
- Service fault statistics on turnaround times.
- Service fault category statistics.
- Service fault statistics on the history of categories of complaints.
- Service fault hot spot visualisations.

Reports should utilise both text-based reporting and graphics.

Stakeholders of the system include:

- Community Residents
- Ward Councillors
- Local Municipality (service delivery departments and communications officers)
- Community Organisations (schools, NGOs, SPCA, neighbourhood watch groups)
- System Administrators (e.g., municipal IT or host institution)

System business rules (please note that these are **SOME** of the business rules and not an exhaustive list; you **should** develop more as a team for your unique system):

- A registered community user or ward councillor is able to create zero or many municipal fault tickets, but each ticket is created by one and only one ward councillor OR community member.
- Each municipal ward can have many fault tickets associated with it, but each ticket is linked to only one ward.
- Each municipal service (water, sanitation, roads, electricity, parks and recreation) can have multiple tickets associated with it, but each ticket pertains to only one municipal service.
- Each ward has a single councillor, and each councillor is associated with a single ward.
- Each ticket has one status associated with it at a time, but each ticket status can have multiple tickets associated at a time.
- Ticket statuses can be changed.
- Ward councillors can add comments to tickets; each comment is associated with one ticket.

- Ward councillors can update the ticket status of a municipal fault ticket.
- The system administrators verify user account roles and add notices to the community space.
- Municipal communications officers can add notices to the community space and track tickets system-wide.
- All users have access to reporting features.

Note: Not all business rules translate into data entities; some specify system permissions, interfaces, and access levels.